

Technical documentation of the Global Transport Costs Dataset for International Trade

Version from June 2023

1. Introduction

This document provides an overview of the content of the enhanced and extended Global Transport Costs Dataset for International Trade (GTCDIT), in its state of June 2023, and describes the sources and methods used for its compilation. The dataset has been built within the initiative “Global Transport Costs Dataset on International Trade and Resulting Analytics” carried out by UNCTAD and the World Bank with the primary aim to extend and enhance the beta version of GTCDIT released on UNCTAD’s online database UNCTADstat in 2020 (Hoffmeister et al., 2022).

The content of the current state of the dataset is described in the next section. The sources used are described in section 4, and the main compilation steps in Section 4.

2. Description of the dataset

The dataset offers primary data on international trade and the costs of its transport from 237 origin countries to 33 destination countries, broken down by 1223 commodity groups and five modes of transport: air, sea, rail, road, and others. The breakdown by mode is available for 26 out of the 33 destination countries, in at least one year. The dataset records international trade flows, in values and volumes, alongside the work and expenditures incurred for their transport, **by pair of trading partners**, commodity and mode of transport by which goods entered the destination country, annually, from 2016 to 2021.

The variables included in the dataset and a description of their meaning and the way they are coded is presented in Table 1.

Table 1. The variables of the dataset

a) Dimensions

Commodity (CommodityProduct)	
Description	The traded commodity
Categories	Commodity groups as defined by the headings (4-digit level) of the Harmonized System (HS) Nomenclature, 2017 edition (World Customs Organization, 2023; UNSD, 2023a). Unidentified commodities are coded by “xx” in the last two digits or by “xxxx” in all digits.
Country of origin (Origin)	
Description	The country of origin as reported by the importer. According to IMTS Concepts and Definitions (UNSD, 2011), this refers to the country in which the traded goods have been produced or manufactured, in accordance with the Revised Kyoto Convention, Specific Annex K / Chapter 1/ E1.
Categories	Economies, as defined by the UNCTAD classification of economies (UNCTAD, 2023). Category “xxx” for unidentified origin countries.

Destination country (Destination)	
Description	The country or territory that has reported an import in its international merchandise trade statistics submitted to the United Nations Statistics Division (UNSD).
Categories	Economies, as defined by the UNCTAD classification of economies (UNCTAD, 2023).
Last mode of transport (LastMoT)	
Description	The mode by which internationally traded goods entered the economic territory of the destination country (<i>Destination</i>), in accordance with IMTS Concepts and Definitions, article 7.1 (UNSD, 2011)
Categories	10 - Air 21 - Sea 31 - Railway 32 - Road 99 - Other modes
Reference year (Year)	
Description	Year in which the internationally trade goods enter the territory of the destination country, usually approximated by the year of the lodgement of the customs declaration (UNSD, 2011)
Categories	2016, 2017, ..., 2021

b) Facts

FOB value (FOBValue)	
Description	The FOB-type value of internationally traded goods, as defined by IMTS Concepts and Definitions, article 4.6 (UNSD, 2011). It includes the transaction value of the goods and the value of services performed to deliver the goods from their origin to the border of the exporting country. It is usually derived by customs administrations from the invoice price and the terms of delivery indicated in the contract of sale, as well as from other supporting documents.
Unit of measure	United States dollars
Weight (QtyKg)	
Description	The weight of the internationally traded goods net of packaging.
Unit of measure	Kilograms
Distance (Distance)	
Description	The average distance over which internationally traded goods were deemed to be transported for their delivery from main city centres of the country of origin (<i>Origin</i>) to main city centres of the destination country (<i>Destination</i>).

Unit of measure	Kilometers
Transport expenditures (TransportCosts)	
Description	The value of services performed to deliver internationally traded goods from the border of the country of origin (<i>Origin</i>) to the border of the exporting country (<i>Destination</i>) including, as main components, the service of shipping the goods and the procurement of insurance against the risk of loss or damage during the carriage. Transport expenditures are calculated as the difference between the CIF value and the FOB value, as defined by IMTS Concepts and Definitions, article 4.6 (UNSD, 2011), building on the definitions in Incoterms (International Chamber of Commerce, 2019). The CIF and the FOB values are usually derived by customs administrations from the invoice price and the terms of delivery indicated in the contract of sale, as well as from other supporting documents.
Unit of measure	United States dollars
Ad-valorem freight rate (TC.Value)	
Description	The ratio of transport expenditures (<i>TransportCosts</i>) to the FOB value (<i>FOBValue</i>) of the transported goods
Unit of measure	Ratio
FreightRate per unit (TC.t)	
Description	The ratio of transport expenditures (<i>TransportCosts</i>) to the physical mass of the transported goods (<i>QtyKg</i>)
Unit of measure	United States dollars per kilogram
Ad-valorem freight rate per kilometer (TC.Value.km)	
Description	The average expenditure (<i>TransportCosts</i>) for the transport of internationally traded goods over a given distance (<i>Distance</i>), as percentage of their FOB value (<i>FOBValue</i>)
Unit of measure	Ratio per 10 000 kilometer
Freight rate per ton-km (TC.t.km)	
Description	The average expenditure (<i>TransportCosts</i>) per ton of internationally traded goods (<i>QtyKg</i>) for their transport over a given distance (<i>Distance</i>)
Unit of measure	United States dollars / ton-kilometer

Is it for ton kilometer or ton of good ?

To avoid misinterpretation, some particularities in the variable definitions require attention:

- a) The transport mode recorded in the variable LastMoT may not be the only mode used during the transport from the country of origin to the destination country. Before reaching the final destination, goods may have been transported by other modes and transloaded on their way from the origin to the destination countries. In the next enhancement of the dataset, an adjustment for transshipment is intended to be implemented.
- b) Since transport costs are derived as the difference between the CIF and the FOB value, they cover more than the pure costs of moving the imported goods geographically. They also include the costs of insurance for losses and damage. These costs are part of the trade bill and, under free competition, can be taken as a measure of the average expected costs caused by those losses and damage.
- c) The transport costs recorded in the database are meant to refer to the transport up to the border of the destination country, whereas distance is measured up to city centres located within that country. Therefore, the ad-valorem freight rate per km and the freight rate per ton-kilometer may be slightly underreported.
- d) As the distance between the origin and the destination country is calculated as an average of the minimum distance needed to transport goods between main city centres of these countries, it should be considered as a first best estimate of the average distance which goods actually travelled. This average does not take into account the intra-country distribution of trade, which is not currently available. **It should also be considered that, for logistic reasons, the direct route is not always the most cost-efficient one. Carriers may decide to let goods travel over a longer distance than technically needed.**

In its current state, the dataset has around 4 million rows and uses 340 MB of memory space when stored in CSV-format. For easier processing and data exchange, it has been split up into 91 files, in the following referred to as trade files, where each trade file records the imports, transport costs and derived indicators of a specific destination country in a specific reference year. To keep file sizes low, the codes of the destination country and the reference year are provided in the file name rather than within the file. The filename follows the convention: “[Destination] _ [Year] _Imports.csv”.

3. Data sources

The dataset has been built from two main sources: UN Comtrade and a distance matrix constructed from geographic information systems (GIS).

3.1. UN Comtrade

UN Comtrade is a well-known and widely recognized comprehensive database on international trade, constructed from international trade data reported by national statistical agencies, mainly based on the customs records presented at country borders, and publicly released by UNSD

(United Nations, 2023). Among other data, the database records the values and volumes of merchandise exports and imports, broken down by reporting country, trading partner and product group. The data obtained from countries are released in UN Comtrade almost in their raw form, after only basic editing and validation checks have been applied.

The 2010 revision of global reporting standards on international merchandise trade statistics (UNSD, 2011) have led to the inclusion of additional details in the countries' data submissions to UNSD. With the "Upgrade 2019", these have been added to UN Comtrade as new variables and breakdowns, opening up new opportunities for the measurement of transport costs detailed by mode. Firstly, while in the conventional UN Comtrade database each flow was recorded on a CIF or FOB basis only – imports usually on a CIF basis and exports on an FOB basis – for some reporting countries both values have now become available for imports. This combined reporting of CIF and FOB values allows us calculating transport costs as the difference between the two (see section 2 above). Secondly, for some countries, trade flows are now broken down by the transport mode by which the goods entered the destination country (UNSD, 2023b).

3.2. Time Distance Matrix

The variable *distance* is compiled from a distance matrix constructed in the scope of this project. This matrix records estimates of the average distance goods need to travel at minimum to transport them from main city centres of one country to main city centres of another using a given mode of transport. For the development of the distance matrix, in the first step, distances computed between 333 origins and destinations at major city level (centroids) were computed using a shortest-path model that identifies the shortest distance between locations, based on a GIS, after endogenizing travel times by type of infrastructure, intermodal transport, dwelling times, and mode-specific travel times (see figure 1). For more information on the construction of the algorithm and the underlying data, see Halim et al. (2018). In a second step, the distances between the centroids of the origin and destination countries were aggregated to distances between countries by calculating the median.

4. Compilation steps

For the compilation of GTCDIT, the UN Comtrade data are retrieved and incorporated into the UNCTAD Statistics data systems, they are harmonized and reformatted to make them better usable for the data output, they are merged with the Time Distance Matrix, and, finally, extended by the calculation of derived variables. These transformation steps are described in more detail in the following sections.

4.1. Extraction of the UN Comtrade data

The UN Comtrade data are retrieved through an API from the UN Comtrade Azur cloud, trade file by trade file, where a trade file represents the data transmission from one reporting country for a given flow (exports or imports) in a given reference year.

These data extractions include the following variables:

- the reference year (*Year*)
- the flow type (*flowCode*) to distinguish between imports, exports, re-imports and re-exports
- the code of the reporting country (*reporterCode*),
- the code of the partner country (*partnerCode*);
- the commodity code (*cmdCode*);
- the product classification used and its version (*classificationCode*)
- the code that identifies the mode of transport by which goods entered or left the reporting country (*motCode*);
- the code that identifies the quantity unit used for the measurement of volumes (*qtyUnitCode*);
- net weight (*netWgt*)
- the information whether net weight is estimated (*isnetWgtEstimated*)
- the CIF value of trade in United States dollars (*CIFValue*)
- the FOB value of trade in United States dollars (*FOBValue*);
- the volume of trade (*qty*)
- the country of consignment (*partner2Code*)
- the customs procedure code (*customsCode*)

The variables *flowCode*, *partner2Code* and *customsCode* are solely used to select the needed records. The other variables are used for further processing. For the variables *reporterCode* and *partnerCode*, a code conversion from M49, the classification of countries and areas used in UN Comtrade (UNSD, 2023c), to the corresponding codes used in UNCTADstat (UNCTAD, 2023) is applied.

The extracted data include the information needed from UN Comtrade to build GTCDIT. However, they are not yet fully comparable across destination countries, commodity groups, and over time, and the way they are formatted is not ideal for later analyses, especially if these involve aggregations. The comparability across destination countries and time is limited firstly by the fact that commodities are coded in slightly different ways across trade files, as different vintages of the Harmonized System (HS) have been applied. Secondly, quantities of different commodities are recorded in different units of measure even within the trade files. The quantity of some commodities is recorded in kg, of others in number of items, liters, meters, carat and other units. This considerably constraints the possibilities to sum the volumes up over commodity groups.

Furthermore, the classification of commodities and last mode of transport is more detailed than needed for the output dataset. With respect to the dimensions trading partner, commodity, and last mode of transport, the dataset includes group totals and subtotals as well as values of individual subitems. Therefore, summing values up by groups can lead to double counting. This can render analyses involving aggregations difficult.

To address the shortcomings above, (1) the extracted UN Comtrade data are harmonized with reference to the classification of commodities, (2) quantities are converted from their original units of measure into kilograms, and (3) totals and sub-totals are dropped, and values of individual categories adjusted to avoid under-reporting.

4.2. Harmonization of commodity codes

To ensure that the data can be reasonably compared and combined with respect to commodity groups, the data from all trade files are converted to correspond with the 2017 Revision of the HS classification, unless they had already been aligned with that version of the classification, using the correspondence tables available on the website of the United Nations Statistics Division (UNSD, 2023a). Depending on the way the categories are linked between different versions, different treatments are applied.

- If the HS code is identical in the original and the target classification (one-to-one match), no conversion is required, and the observation value is left as is.
- If a single code in the original classification is associated with multiple codes in the target classification (one-to-many match), the treatment depends on whether the reporting country has ever published in HS-2017 in other years or not. If it has, the single observation value from the original classification is split up to the codes of the target classification in the same proportions as observed in the country's HS-2017 data from other years. If it has not, or the product code is not present in the country's HS-2017 data, equal allocation is applied.
- If multiple codes in the original classification are associated with a single code in the target classification (many-to-one match), the observation values of these multiple codes are summed up.
- If multiple codes in the source classification are associated with multiple different codes in the target classification (many-to-many match), again, the treatment depends on whether the reporting country has ever published in HS-2017 in other years or not. If it has, the summed observation values from the original classification are split up to the codes of the target classification in the same proportions as observed in the country's HS-2017 data from other years. If it has not, or the product code is not present in the country's HS-2017 data, equal allocation is applied.

4.3. Conversion of quantity into net-weight

Although countries are recommended to include information on the net weight of the imported and exported goods in their trade data submitted to UN Comtrade, in addition to the information on quantity expressed in a different unit of measure, and despite efforts undertaken by the United Nations Statistics Department to fill these gaps using estimations, the values of net weight have been found to be missing in 19 per cent of the cases throughout the extracted UN Comtrade data. To obtain approximately full data coverage in the variable *QtyKg*, we estimate those missing values by calculating standard conversion factors (ρ), differentiated by group of commodity (g), original unit of measure (Φ), and, if available, last mode of transport (m), based on the values observed in the data, and we multiply the quantities reported by countries in the original unit of measure (q) by those factors:

$$(1) \quad \hat{q}_{t,g,o,d,m}^{kg} = \hat{\rho}_{\Phi,g,m} q_{t,g,o,d,m}$$

where o is the origin country and d the destination country of the observation for which the missing value is imputed.

The standard conversion factors are calculated as the median ratio of net weight in kg to the quantity expressed in the original unit of measure, calculated over all observations found for a given combination of commodity group, original unit of measure, and last mode of transport.

$$(2) \quad \hat{\rho}_{\Phi,g,m} = \text{median} \left(\frac{q_{t,g,o,d,m}^{kg}}{q_{\Phi,t,g,o,d,m}} \right)$$

4.4. Reconciliation of totals, subtotals and values of individual categories

To convert the coding of origin and destination countries, oriented on

In UN Comtrade, the identification of commodity groups, trading partners and last mode of transport in UN Comtrade follows a nested structure, in the sense that it contains rows for totals and, in the case of the commodity breakdown, subtotals, as well as values of individual subcategories. In the standardized dataset, we want all values to refer to the lowest intended level of disaggregation: commodity groups defined at heading level (4 digit-codes) of HS; individual partner countries, as opposed to the total world as partner; in individual modes of transport, as opposed to the transport mode total. That way, the data can be summed up in all possible ways across the different dimensions, without any risk of double-counting.

A complication arises from the fact that not all trade is classified by commodity group at the highest level of detail in UN Comtrade, for example for confidentiality reasons. Some trade may be considered in the aggregate for a chapter of the classification (identified by 2-digit HS codes) but may not be separately shown at the level of headings (identified by 4-digit HS codes). It may even be the case that it is considered in the commodity total but not separately reported at the level of chapters or headings. Likewise, the trading partner of some trade may not be separately reported but included in the aggregate of the trade with the whole world.

Before deleting any totals and subtotals from the dataset, we therefore calculate residual categories, with respect to commodity or trading partner, that cover the trade that is not separately reported in UN Comtrade or cannot be assigned to any target category for other reasons. Let Γ denote the set of the one hundred HS chapters, represented by the 2-digit code i , and Γ_i denote the set of headings j , identified by 4-digit codes, within chapter i . Let the subscripts “ $_$ ” denote the overall total and “ $_{i_}$ ” the subtotal of chapter i . We calculate the value of variable x in the corresponding residual categories as

$$(3) \quad x_{i,*} = x_{i_} - \sum_{j \in \Gamma_i} x_{i,j}$$

and

$$(4) \quad x_* = x_- - \sum_{i \in \Gamma} x_i$$

These residual categories are identified in the output data by “xx” in the last two digits of the commodity code or by “xxxx” in all digits of the code. That way, the totals x_- and $x_{i_}$, which we drop, can always be re-established by summing up all observations within Γ or Γ_i , respectively, including the residual categories x_* and $x_{i,*}$.

Likewise, unidentified countries of origin, derived from the UN Comtrade variable on the partner country (*partnerCode*), are assigned to a residual category coded as “xxx”. Modes of transport other than air (“10”), sea (“21”), rail (“31”) and road (“32”) are assigned to a residual category with code “99”, using the same method.

4.5. Calculation of derived variables

At the end, the so harmonized and reformatted UN Comtrade data are merged with the Time Distance Matrix, to be able to combine them with the information on distance and to construct the target values of GTCDIT. While the FOB value (FOBValue), the weight in kilogram (QtyKg) and distance (Distance) are directly observed in the input data, transport expenditures

(TransportCosts), the ad-valorem freight rate (TC.Value), the freight rate per unit (TC.t), the ad-valorem freight rate per kilometer (TC.Value.km) and the freight rate per ton-km (TC.t.km) need to be calculated. The formulas for these calculations are given below:

Transport expenditures:

$$(5) \quad \text{TransportCosts} = \text{CIFValue} - \text{FOBValue}$$

Ad-valorem freight rate:

$$(6) \quad \text{TC.Value} = \frac{\text{TransportCosts}}{\text{FOBValue}}$$

Freight rate per unit:

$$(7) \quad \text{TC.t} = \frac{\text{TransportCosts}}{\text{QtyKg}}$$

Ad-valorem freight rate per kilometer:

$$(8) \quad \text{TC.Value.km} = \frac{\text{TransportCosts}}{\text{FOBValue} \times \text{Distance}}$$

Freight rate per ton-kilometer:

$$(9) \quad \text{TC.t.km} = \frac{\text{TransportCosts}}{\text{QtyKg} \times \text{Distance}}$$

5. Outlook

As mentioned above, the present version of the enhanced and extended GTCDIT covers only the merchandise trade for which transport costs can be directly observed in the UN Comtrade data, where transshipment is not taken into account. It is envisaged, however, to develop a dataset with approximately global coverage, that provides reasonable estimates of the actual use of the different transport modes for international trade, not just the mode by which goods entered the destination country. For this, the project team is currently carrying out the necessary imputations to fill data gaps and to estimate modal split, based on estimation models and a refined Time Distance Matrix developed during the project. This refined version of GTCDIT is intended to be released by June 2024.

References

- Halim RA, Kirstein L, Merk O and Martinez L (2018). Decarbonization pathways for international maritime transport: A model-based policy impact assessment. *Sustainability*. 10(7):2243.
- Hoffmeister O et al. (2022). Developing a global transport costs dataset for international trade. UNCTAD Research Paper No. 85. (accessed 24 February 2023).
- International Chamber of Commerce (2019). *Incoterms 2020*.
- UNCTAD (2023). UNCTADstat: Classifications. Available at <https://unctadstat.unctad.org/EN/Classifications.html> (accessed 1 October 2023).
- United Nations (2023). UN Comtrade database. Available at <https://comtrade.un.org/> (accessed 10 November 2023).
- UNSD (2011). International Merchandise Trade Statistics: Concepts and definitions 2010. ST/ESA/STAT/SER.M/52/Rev.3. United Nations publication. Sales No. E.10.XVII.13.
- UNSD (2023a). Classifications on economic statistics. Available at <https://unstats.un.org/unsd/classifications/Econ> (accessed 8 November 2023).
- UNSD (2023b). Methodology guide for UN Comtrade user on UN Comtrade Upgrade 2019. Available at <https://comtrade.un.org/data/doc/UpgradePlan> (accessed 13 November 2023).
- UNSD (2023c). Standard country or area codes for statistical use (M49). Available at <https://unstats.un.org/unsd/methodology/m49/> (accessed 27 November 2023).
- World Customs Organization (2023). HS Nomenclature 2017 Edition. Available at <https://www.wcoomd.org/en/topics/nomenclature/instrument-and-tools/hs-nomenclature-2017-edition.aspx> (accessed 10 November 2023).